Quizrr Chapter-Wise Test for JEE Main - 2025

Q.1

The correct order of acidic strength is

(A) $Cl_2O_7 > SO_3 > P_4O_{10}$

(B) $CO_2 > N_2O_4 > SO_3$

(C) $Na_2O > MgO > Al_2O_3$

(D) $K_2O > CaO > MgO$

Q.2

Ammonia, on reaction with hypochlorite anion, can form

(A) NO

(B) NH_4Cl

(C) N_2H_4

(D) HNO_2

Q.3

An oxide of nitrogen is reddish brown and paramagnetic at room temperature but it decolourises and also loses its paramagnetism on freezing it. The oxide at room temperature is

(A) Pure NO_2

(B) Pure N_2O_4

(C) equilibrium mixture of N_2O_4 and NO_2

(D) N_2O_5

Q.4

Which compound acts as an oxidising as well as reducing agent

(A) SO_2

(B) H_2SO_4

(C) Al_2O_3

(D) $K_2Cr_2O_7$

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Q.5

The correct order of pseudohalide, polyhalide and interhalogen are

(A) BrI_2^-, OCN^-, IF_5

(B) IF_5, BrI_2^-, OCN^-

(C) OCN^-, IF_5, BrI_2^-

(D) OCN^-, BrI_2^-, IF_5

Q.6

Sulphuric acid reacts with PCl_5 to give

(A) Thionyl chloride

(B) Sulphur monochloride

(C) Sulphuryl chloride

(D) Sulphur tetrachloride

Q.7

$(NH_4)_2\ Cr_2O_7$ on heating liberates a gas. The same gas will be obtained by

(A) Heating NH_4NO_2

(B) Heating NH_4NO_3

(C) Treating H_2O_2 with $NaNO_2$

(D) Treating $Mg_3\ N_2$ with H_2O

Q.8

Which of the following statements is incorrect $-$

(A) O_3 is used as disinfectant

(B) NO_2 is oxidised to N_2O_5 by O_3

(C) $O_3{}^-$ is paramagnetic in nature

(D) Dry iodide reacts with ozone and form I_2O_5

Q.9

Which of the following allotropic forms of phosphorus is thermodynamically most stable.

(A) White phosphorus

(B) Black phosphorus

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(C) Red phosphorus

(D) All are equally stable

Q.10

Which oxide of the nitrogen exist in two different forms, which can be interconverted by irradiation with light of the appropriate wavelength

(A) NO_2

(B) N_2O_3

(C) N_2O_5

(D) N_2O

Q.11

The major compound from which iodine is isolated is

(A) NaI

(B) $NaIO_3$

(C) $NaIO_4$

(D) I_2O_5

Q.12

An inorganic halide (A) reacts with water to form two acids (B) and (C). (A) also reacts with $NaOH$ to form two salts (D) and (E) which remain in solution. The solution gives white precipitate with both $AgNO_3$ and $BaCl_2$ solutions. What is (A)?

(A) $PCl_3\ F_2$

(B) $SbCl_3$

(C) $AlCl_3$

(D) SO_2Cl_2

Q.13

The most powerful oxidant among the following is

(A) $HOCl$

(B) $HClO_2$

(C) $HClO_3$

(D) $HClO_4$

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Q.14

XeF_6 on hydrolysis gives

(A) XeO_3

(B) XeO

(C) XeO_2

(D) Xe

Q.15

Which reaction is not valid

(A) $HCl + F_2 \rightarrow HF + Cl_2$

(B) $HF + Cl_2 \rightarrow F_2 + HCl$

(C) $Zn + HCl \rightarrow ZnCl_2 + H_2$

(D) $Al + HCl \rightarrow AlCl_3 + H_2$

Q.16

Which one is the anhydride of $HClO_4$

(A) Cl_2O

(B) ClO_2

(C) Cl_2O_6

(D) Cl_2O_7

Q.17

The correct order of the thermal stability of hydrogen halides $(H - X)$ is

(A) $HI > HBr > HCl > HF$

(B) $HF > HCl > HBr > HI$

(C) $HCl < HF < HBr < HI$

(D) $HI > HCl < HF < HBr$

Q.18

When sulphur is boiled with Na_2SO_3 solution, the compound formed is

(A) Sodium sulphide

(B) Sodium sulphate

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(C) Sodium persulphate

(D) Sodium thiosulphate

Q.19

$P_4 + 3NaOH + 3H_2O \rightarrow A + 3NaH_2PO_2$ here, ' A' is

(A) NH_3

(B) PH_3

(C) H_3PO_4

(D) H_3PO_3

Q.20

Bromine is liberated when an aqueous solution of potassium bromide is treated with

(A) Cl_2

(B) I_2

(C) Dilute H_2SO_4

(D) SO_2

Q.21

What is the oxidation number of phosphorous in orthophosphoric acid?

Q.22

The $F - Xe - F$ bond angle in XeF_4 is

Q.23

The number of paired electrons around Xe in XeO_3 is

Q.24

How many P-O-P bonds are present in P_4O_8?

Q.25

How many electrons are present in the valence shell P in PCl_3?

Q.26

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How many S-S bonds are present in S_8 molecule?

Q.27

In the following oxyanions:
i. SO_4^{2-}
ii. $S_2O_5^{2-}$
iii. $S_2O_7^{2-}$
iv. $S_3O_6^2$
If X = Number of oxyanions having two equivalent $S-O_{b_{Ond}}$ per central atom.
Y = Number of oxyanions having three equivalent $S-O_{b_{0ndg}}$ per central atom.
Z = Number of oxyanions having four equivalent $S-O_{b_{0Ond}}$ per central atom.
Find the value of $X+Y+Z$.

Q.28

Sodium iodate is treated with calculated amount of sodium bisulphite to prepare iodine. How many moles of sodium bisulphite are required to prepare one mole of iodine from sodium iodate?

Q.29

In a given sample of bleaching powder, the percentage available chlorine is 49 .The volume in litres of chlorine obtained if $20\ g$ of the sample is treated with HCl_{at} N is.

Q.30

How many $d\pi-p\pi$ bonds are there in XeO_4?

Answers & Solutions

Q.1 Answer:

$Cl_2O_7 > SO_3 > P_4O_{10}$

Solution:

The acidic strength of an acid is defined as the tendency of the acid to dissociate into its corresponding ions in order to release a proton and the anion. The acidic strength of a strong acid is high and that of a weak acid is low. This means that on dissociation, a strong acid releases its proton at a much faster rate than the weak acid.

Oxidation number: The greater the oxidation number of the central atom of the oxide, the greater the acidic strength will be.

In a periodic table, the acidic strength of an atom increases across a period and decreases down the group. In the above options, the correct order of acidic strength of the oxides is:

$Cl_2O_7 > SO_3 > P_4O_{10}$

The oxidation state of the oxides is as follows:

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Let the oxidation number of the chlorine atom be ' x'.

$\mathrm{Cl_2O_7} = 2x + 7(-2) \rightarrow x = +7$

Let the oxidation number of the sulfur atom be $'y'$.

$SO_3 = y + 3(-2) \Rightarrow y = +6$

Let the oxidation number of the phosphorus atom be $'z'$.

$P_4O_{10} - 4z + 10(-2) \rightarrow z - +5$

Thus, as we can see that the order of oxidation is in a decreasing manner, this means that this option is correct.

So, the correct answer is Option (a).

Q.2 Answer:

$\mathrm{N_2H_4}$

Solution:

$$2\mathrm{NH_3} + \mathrm{OCl^-} \rightarrow \mathrm{NH_2 \cdot NH_2} + \mathrm{H_2O} + \mathrm{Cl^-}$$

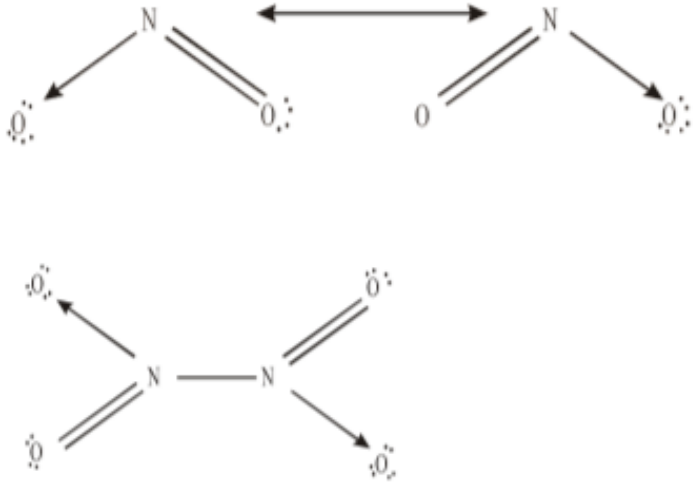
Q.3 Answer:

Pure $\mathrm{NO_2}$

Solution:

$$2\mathrm{Pb\,(NO_3)_2} \rightarrow 2\mathrm{PbO} + 4\mathrm{NO_2} + \mathrm{O_2}$$

i.e. can be prepared by heating heavy metal nitrate of $\mathrm{Pb\,(NO_3)_2}$. It is a red brown gas but on cooling below $273\ \mathrm{K}$ converts to colourless liquid and retains as dimer. In $\mathrm{NO_2}$ molecules 17 valence $\mathrm{e^-}$ are there (12 that of O & 5 that of N) i.e. one $\mathrm{e^-}$ unpaired results to paramagnetic nature. So this is an odd $\mathrm{e^-}$ molecule on being dimer with even $\mathrm{e^-}$ becomes stable & loses paramagnetic characteristics.



Q.4 Answer:

$\mathrm{SO_2}$

Solution:

The minimum and maximum oxidation number of S are -2 and $+6$ respectively. Since the oxidation number of S in $\mathrm{SO_2}$ is $+4$, therefore it can be either increased or decreased. Therefore

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SO_2 **behaves both as an oxidising as well as reducing agent.**

Q.5 Answer:

OCN^-, BrI_2^-, IF_5

Solution:

Some complex ions of nitrogen show similarities with halide ions. These are called pseudohalide ions e.g., NCO^-, CN^- **etc.**

Halogen, among themselves, form complex ions which are called polyhalide ions. e.g., I_3^-, BrI_2^- **etc. Similarly, interhalogens are the compounds of halogen in which one halogen is cation and other halogen is anion. E.g.,** IF_7, ICl_5, BrF_3, IF_5 **etc .**

Q.6 Answer:

Sulphuryl chloride

Solution:

$$HO-SO_2-OH+PCl_5 \rightarrow Cl-SO_2-O \quad +POCl_3+HCl$$

$$HO-SO_2-OH+2PCl_5$$

$$\rightarrow \underset{\text{Sulphuryl chloride}}{Cl-SO_2-Cl}+2POCl_3+2HCl$$

Q.7 Answer:

Heating NH_4NO_2

Solution:

$$(NH_4)_2\,Cr_2O_7 \xrightarrow{\Delta} N_2\uparrow +Cr_2O_3+4H_2O$$

$$NH_4NO_2 \xrightarrow{\Delta} N_2\uparrow +2H_2O$$

Q.8 Answer:

Dry iodide reacts with ozone and form I_2O_5

Solution:

(A) Correct statement

(B) $2NO_2+O_3 \rightarrow N_2O_5+O_2$

(C) O_3^- **is paramagnetic**

(D) $2I_2+9\,[O_3] \rightarrow I_4O_9$ **(yellow solid)** $+9O_2$

Q.9 Answer:

Black phosphorus

Solution:

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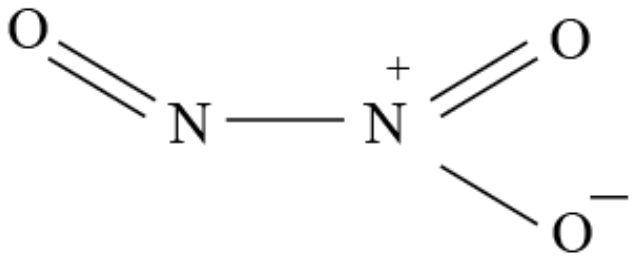
The lattice of black phosphorus is an interlinked six-membered ring in which each atom is bonded to three other atoms. Thus, the reactivity of black phosphorus is very less. Being most stable, black phosphorus is less reactive in nature. Thus, the most thermodynamically stable allotropic form of phosphorus is black.

Q.10 Answer:

N_2O_3

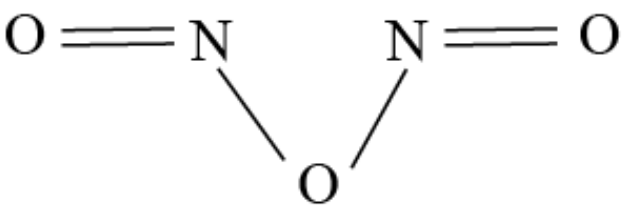
Solution:

N_2O_3 **exists as**



Assymetric form

and



Symetric form

Q.11 Answer:

$NaIO_3$

Solution:

The major compound formed from I_2 isolated as $NaIO_3$

$NaIO_3 + H_2SO_4 \rightarrow NaHSO_4 + HIO_3, HIO_3 + 5HI \rightarrow 3H_2O + 3I_2$

Q.12 Answer:

SO_2Cl_2

Solution:

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$$PCl_3\ F_2 + H_2O \rightarrow H_3PO_4 + HCl + HF$$
$$SbCl_3 + H_2O \rightarrow SbOCl + HCl$$
$$AlCl_3 + H_2O \rightarrow Al(OH)_3 + HCl$$
$$SO_2Cl_2 + H_2O \rightarrow H_2SO_4 + HCl$$
$$H_2SO_4 + BaCl_2 \rightarrow BaSO_4(s)$$

Q.13 Answer:

$HOCl$

Solution:

Acidic strength : $HOCl < HClO_2 < HClO_3 < HClO_4$

Oxidising power: $HOCl > HClO_2 > HClO_3 > HClO_4$

As the stability of oxoanion increases, its tendency to decompose to give O_2 decreases and hence its oxidising power decreases.

Stability of oxoanion:

$CiO^{\Theta} < ClO_2^{\Theta} < ClO_3^{\Theta} < ClO_4^{\Theta}$

Oxidising power: $HOCl > HClO_2 > HClO_3 > HClO_4$.

Q.14 Answer:

XeO_3

Solution:

$$XeF_6 + 3H_2O \rightarrow XeO_3 + 6HF$$

Q.15 Answer:

$HF + Cl_2 \rightarrow F_2 + HCl$

Solution:

Fluorine is more electronegative than chlorine hence, it can displace Cl from HCl while chlorine cannot displace fluorine from HF. Therefore the following reaction is not valid.

$$HF + Cl_2 \rightarrow F_2 + HCl$$

Q.16 Answer:

Cl_2O_7

Solution:

$$2HClO_4 \rightarrow H_2O + Cl_2O_7$$

Q.17 Answer:

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$HF > HCl > HBr > HI$

Solution:

As we move down the group in hydrogen halides the thermal stability decreases due to decrease in bond strength. So the correct order of thermal stability is $HF > HCl > HBr > HI$

Hence, option (b) is correct.

Q.18 Answer:

Sodium thiosulphate

Solution:

When sulphur is boiled with $Na_2S_2O_3$ solution, the compound formed is sodium thiosulphate.

$8Na_2S_2O_3 + S \rightarrow 8Na_2S_2O_3$

Hence, option (d) is correct.

Q.19 Answer:

PH_3

Solution:

This is the laboratory method of preparing phosphine gas.

$$P_4 + 3NaOH + 3H_2O \rightarrow PH_3 + 3NaH_2PO_2$$

In this reaction, phosphorus disproportionate into phosphine and sodium hydrogen phosphite.

Q.20 Answer:

Cl_2

Solution:

$$Cl_2 + 2KBr \rightarrow 2KCl + Br_2$$

A more electronegative halogen can displace less electronegative halogen.

Q.21 Answer:

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Solution:

(5)

Orthophosphoric acid (H_3PO_4)

$3 + x - 8 = 0 \Rightarrow x = 5$

Q.22 Answer:

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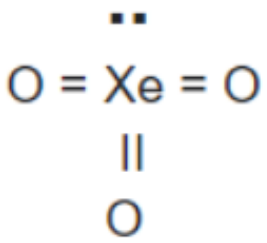
Solution:

The $F-Xe-F$ bond angle in XeF_4 is 90°. In XeF_4 molecule the Xe is at the centre and fluorine atoms are directed towards four corners of a square. The geometry is square pyramidal.

Q.23 Answer:

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Solution:



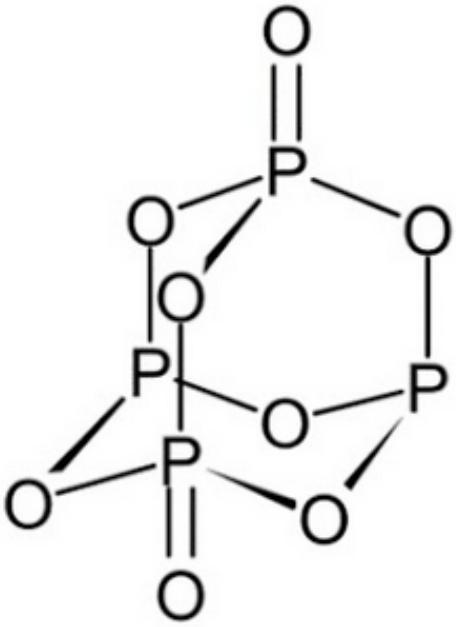
1 lone pair and 3 bond pair ; Total 4 Paired electrons

Q.24 Answer:

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Solution:

P_4O_8 molecule contains six POP bonds.

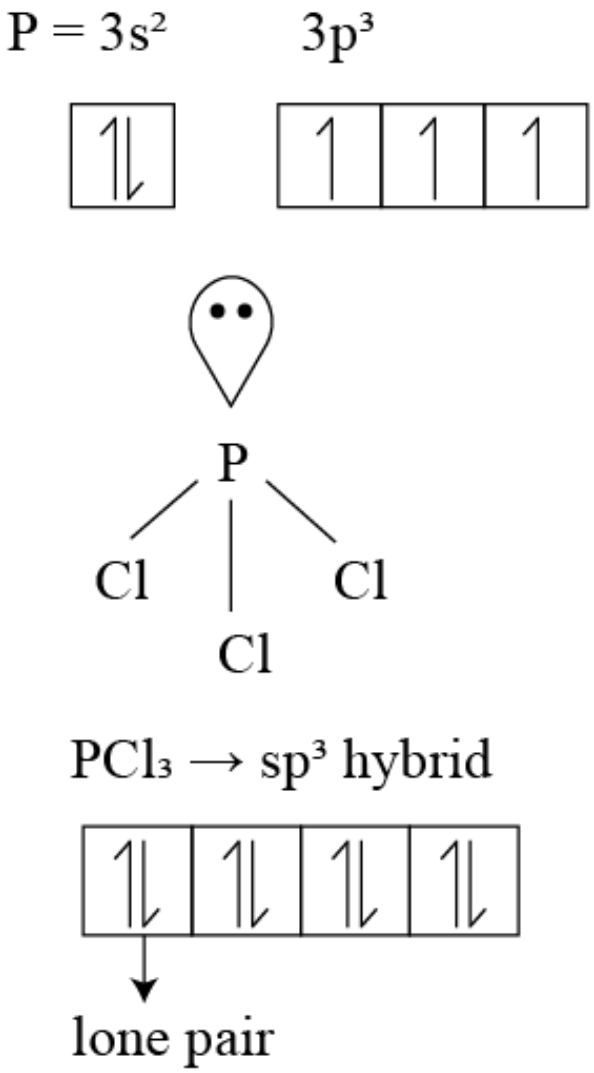


Q.25 Answer:

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Solution:

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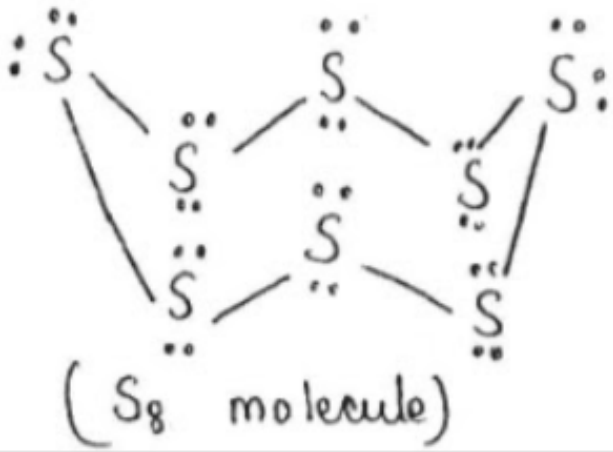
Therefore, number of electrons present in valence shell of P In $pd_3 = 8$

Q.26 Answer:

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Solution:

As atoms in S_8 molecule undergo sp ^{3} hybridisation and contain two lone pair of electrons on each and exists as staggered 8 atom-rings It has a cuown structure.

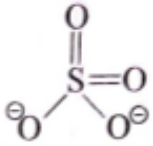
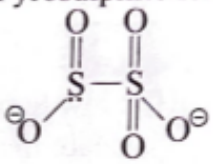
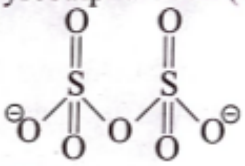
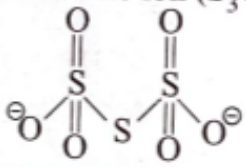


Q.27 Answer:

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Solution:

Oxyanions Sulphate ion (SO_4^{2-})	No. of equivalent bonds
i. 	X = 2
ii. Pyrosulphite ion ($S_2O_5^{2-}$) 	X = 2 Y = 3
iii. Pyrosulphate ion ($S_2O_7^{2-}$) 	Y = 3
iv. Trithionate ion ($S_3O_6^{2-}$) 	Y = 3
X + Y + Z = (2 + 2) + (3 + 3 + 3) + 0 = 12	

Q.28 Answer:

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Solution:

From above balanced equation it is clear that 5 moles of sodium bisulphite gives 1 mole of I_2

$$2NaIO_3 + \underset{5\,\text{mole}}{5NaHSO_3} \longrightarrow 3NaHSO_4 + 2Na_2SO_4 + H_2O + \underset{\text{one mole}}{I_2}$$

Q.29 Answer:

—

Solution:

Weknow, bleaching powder conteins $49\%\mathrm{Cl}$.

mole $= \frac{49}{100} \times \text{mas} \times \frac{1}{\text{melare mass}}$

valume $= \frac{49}{100} \times 20 \times \frac{1}{71} \times 22.4L/\text{mol} = 3\text{ L}$ $S_0, 20$ g **af** $\mathrm{NaO}U_2$ **will gine** 3 L **a** Cl_2.

Q.30 Answer:

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Solution:

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Structure of XeO_4 has $4p\pi - d\pi$ bonds. Here, *p*-orbital is from O and d-orbital is from Xe - atom respectively.

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